

3

To celebrate the Millennium in the year 2000, a footbridge was constructed across the River Thames in London. After the bridge was opened to the public it was discovered that the structure could easily be set into oscillation when large numbers of pedestrians were walking across it.

- (i) State the name of the phenomenon described above.

..... [1]

- (ii) Explain why this phenomenon can become hazardous.

.....

 [3]

- (iii) Engineers looked at the situation and applied more damping to the bridge.

Fig. 3.1 shows the variation with driving frequency of the amplitude of the oscillation of the bridge after applying the damping.

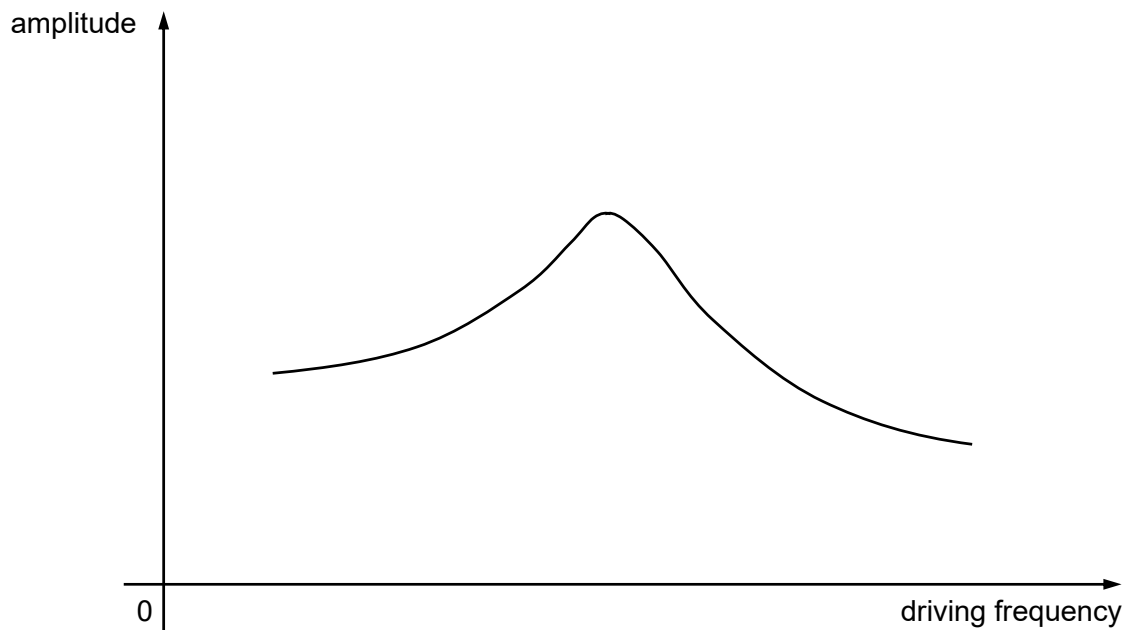


Fig. 3.1

On Fig. 3.1, sketch the graph of the variation with driving frequency of the amplitude before damping was applied. [2]

[Total: 6]